

'Quantum Supremacy' with DST for building the country's own fifty-qubit quantum computer in the next four to five years. DST has already started working on a draft for the mission.

DST is also working on a major security application of quantum computing. This is related to RSA encryption—one of the most widely used encryption standards for secure communications that prevent third-parties from reading your private messages.

Breaking this encryption requires being able to perform billions of calculations and breaking down large numbers into their prime factors. This is precisely what quantum computers will be good at. In other words, the prime factorisation quantum algorithm can be potentially applied to overcome RSA encryption.

Once quantum technology goes mainstream, the most secure encryptions generated using today's crypto algorithms could be broken into within a matter of seconds—something governments worldwide are scared about. Quantum changes computing in the most fundamental way since it provides exponentially more compute power.

Other countries

Australia's Quantum Computing Academy consists of a series of five two-week, online, hands-on, learning workshops where school students explore actual quantum computers, run specialised algorithms, and learn about quantum cryptography and quantum networking. Generous scholarships are available for eligible students.

The Quantum Learning Machine (QLM) of Japan is a specially config-

ured classical computer designed to simulate quantum algorithms using up to forty qubits.

The UK government has recently provided seven and a half million pounds of grants for the development of a new quantum operating system called Deltaflow.

Conclusion and suggestions

As a nation, we have computer architects and quantum information theorists. We have more people on the software side, and hardly any hardware person in the quantum computing area. However, we need all these people and also those running the electronics industry to translate the raw data from theoretical modelling into hardware.

Quantum computers are the next big step in computing technology. These use the fundamentals of quantum mechanics to process significantly greater amounts of information much faster than classical computers. It is expected that when the error-corrected and fault-tolerant quantum computation is built, scientific and technological advancements will occur at an unprecedented scale.

Quantum computers are very urgently needed in India to tackle large scale problems such as the creation of new medicines, complex cryptography, space applications, and many more. The suggested road map for acquiring and mastering the growing quantum technology is summarised below:

1. Establish an independent National Quantum Computing Academy (NQCA) under the Department of Science & Technology with active industry participation.

2. The academy will be headed by a chairperson, assisted by a board of directors on corporate lines.

3. The academy will oversee and guide all activities covering next-generation technologies, including three segments—quantum computing, quantum communications, and quantum sensing and measurement. The academy will also draw up a national quantum technology strategy for attracting and retaining quantum talent, exploring funding mechanisms, and assessing industry capabilities and infrastructure facilities in close consultation with stakeholders.

4. NQCA will be responsible for building India's own fifty-qubit quantum computer.

5. NQCA will establish hands-on learning workshops on a regular basis for high school students to teach them about quantum technology and further promote deeper learning on quantum computing at the university degree level and finally at the post-graduate and doctoral level to meet the required qualified technical personnel.

6. NQCA will identify appropriate electronics industry for quantum computing and associate with them from the beginning so as to establish a firm industrial base for hardware development and ultimate production sources.

7. Establish an exclusive quantum modern lab facility for both development and in-house manufacturing for QCI's quantum computer activities based upon their superconducting technology. The lab will also enable researchers to reduce the gap between research and product development. This would be complementary to the existing excellent lab already in use at TIFR, Mumbai. **EFY**



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